

Infrared OBJECT DETECTOR

(Infrared Images Object Detection in Realtime)



What was Used?



1. Object Detection with Deep Learning
2. FLIR thermal dataset
3. AI night Object Detection using **pytorch** and **YOLOv5**

Object Detection with Deep Learning

Yolo - You only look once , offers speed and accuracy. We will apply transfer learning - consist of pretraining and finetuning. Yolo detector retrained on Microsoft COCO dataset , fine tune on custom dataset - Thermal imaging dataset



Discovering the FLIR thermal Starter imaging dataset

Our custom dataset is thermal imaging dataset - objects at night,

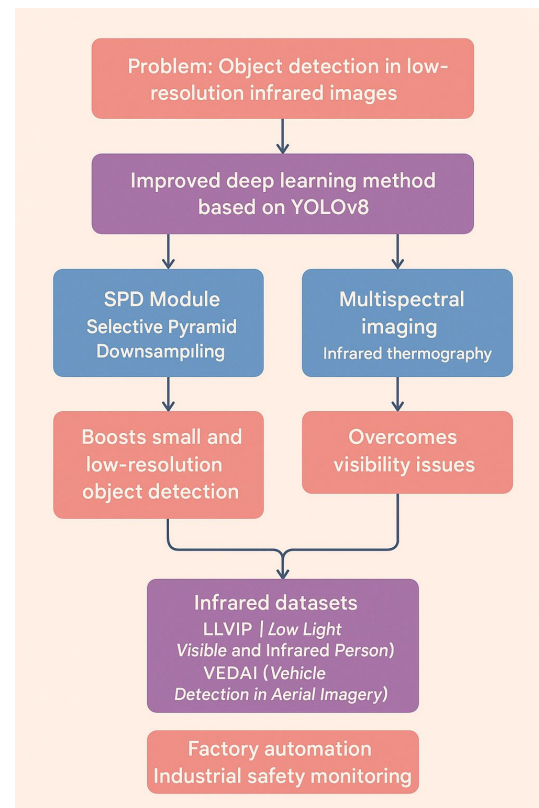
Dataset - Teledyne FLIR ADAS Dataset - consists of more than **14000** thermal images in Grey and Grey 16 format

This can detect 4 different objects that are already hand labelled - **Car, Person, Bicycle, Dog**

To simplify we only use 1772 images , **Training : 60%**, **Validation: 40%**, 1000 images Training , 772 Validation

Review of paper

- ❖ **Problem Addressed:**
Object detection in low-resolution infrared images, especially under challenging conditions like fog, smoke, and low light in factory environments.
- ❖ **Proposed Solution:**
An improved deep learning method based on the YOLOv8 architecture, enhanced for better infrared image detection.
- ❖ **Key Innovation – SPD Module:**
Introduces a **Selective Pyramid Downsampling (SPD)** module.
Boosts detection accuracy for small and low-resolution objects.
Increases inference speed through efficient feature map processing.
- ❖ **Use of Multispectral Imaging:**
Leverages **infrared thermography** to overcome visibility issues.
Enhances overall object recognition in harsh visual environments.
- ❖ **Datasets Used:**
LLVIP (Low-Light Visible and Infrared Person)
VEDAI (Vehicle Detection in Aerial Imagery)
- ❖ **Results:** The proposed model outperforms standard YOLOv8 in detection accuracy and speed on infrared datasets.
Shows significant improvement in identifying small and hard-to-detect objects.



YOLO-FIRI: Improved YOLOv5 for Infrared Image Object Detection

The authors develop **YOLO-FIR** (base) and **YOLO-FIRI** (improved), both based on **YOLOv5**, optimized for IR detection.

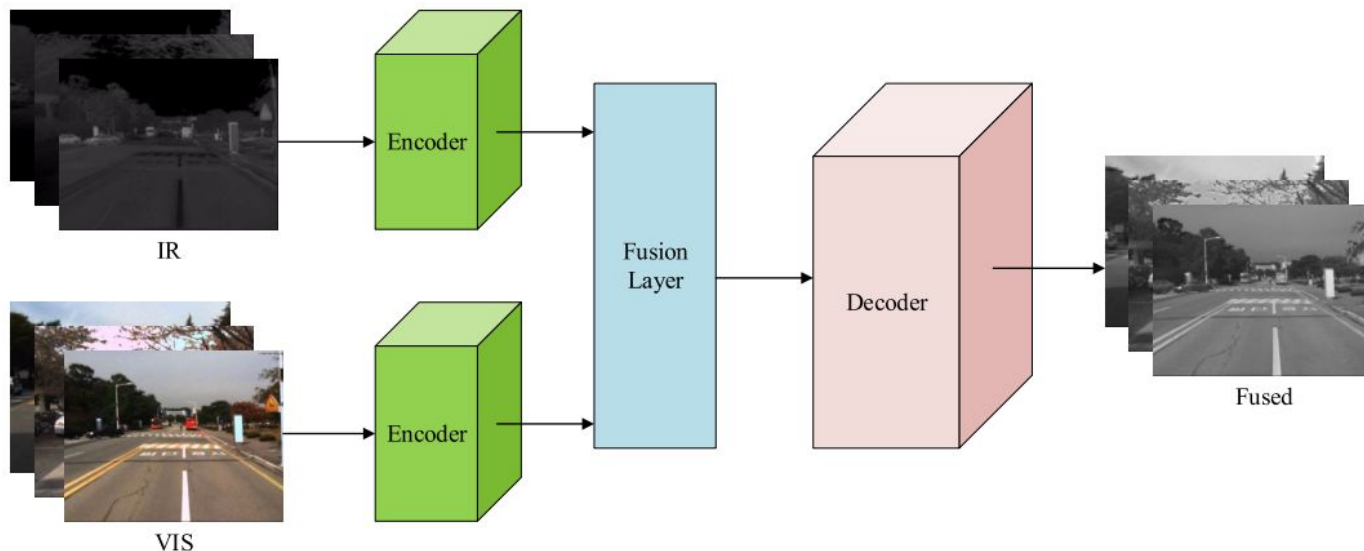


FIGURE 9. The architecture of the Densefuse network. The input images are the well-aligned visible and infrared images. The encoder network uses a convolutional and dense block to extract more useful features from the source images. The fusion layer is used to fuse the extracted features. The decoder network can rebuild the fused infrared images to realize the fused KAIST dataset.

Results

- **Datasets:** KAIST (IR pedestrian), FLIR (multi-class IR).
- **Performance Gains:**
 - On **KAIST**:
 - mAP50 increased by **~21%** over YOLOv4.
 - Detection accuracy (AP, AR, F1) improved **4–13%** over YOLO-FIR baseline.
 - Weight size reduced by **94%**, FLOPs reduced by **84%**, enabling mobile use.
 - On **FLIR**:
 - YOLO-FIRI reached **83.5% mAP**, outperforming ThermalDet and Faster R-CNN baselines

Key Concepts from YOLO-FIRI

- **Extended CSP modules** → stronger shallow feature extraction for weak IR signals
- **Improved SK Attention** → suppress noisy backgrounds, highlight objects
- **Multiscale Detection Head** → adds 4th detection scale for small & distant targets
- **Image Fusion (DenseFuse)** → combine IR + visible images for richer features

Challenges in Real-World IR

- Limited labeled IR datasets (mostly **KAIST, FLIR, OTCBVS**)
- Objects beyond pedestrians/vehicles are underrepresented
- High noise, low resolution, and domain shift vs. RGB

Takeaway

- YOLO-FIRI demonstrates how **attention + multiscale + fusion** dramatically improve IR detection.
- By combining these methods with **transfer learning and creative data strategies**, we can train robust detectors for **new IR objects in real-world environments**.

Practical Extensions

1. Transfer Learning

- Fine-tune pretrained YOLOv8 (COCO weights) on limited IR data

2. Data Augmentation

- CLAHE, histogram equalization, brightness/contrast, Gaussian noise
- Synthetic IR via **GANs/Style Transfer** from RGB datasets

3. Fusion-Based Training

- Capture **RGB + IR pairs**, fuse to generate enhanced training data

4. Expand Object Classes

- Extend detection beyond pedestrians → animals, drones, machinery, etc.
- Collect small custom IR datasets with thermal cameras

5. Deployment & Testing

- Real-time IR streams (drones, CCTV, autonomous systems)
- Evaluate not just mAP, but **false alarm rate & speed on edge devices**